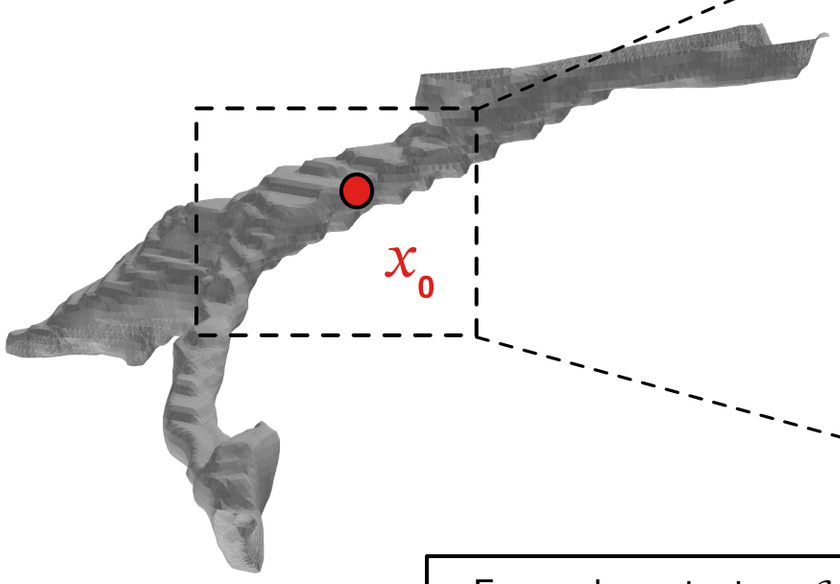
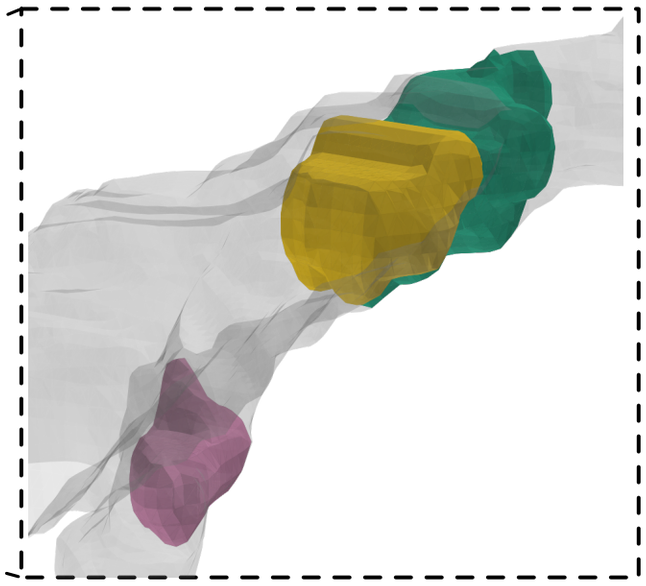
Falsely merged object



Candidate native contacts $\mathcal{C}$
(centroids within R = 500 nm of x_0)



For each contact $e \in \mathcal{C}$: pin endpoint supervoxels to opposing terminals → seeded s-t min-cut → induced native partition P_e

CC-Auto

Automatic single-contact selection

Connectivity screening
Two connected sides, if available; otherwise, fewest extra components

Partition features & normalization
Compute $f_s(P_e)$ and $f_c(P_e)$ for $e \in \mathcal{T}$; median/MAD normalize within $\mathcal{T}$, clipped to ± 8

Best connectivity tier $\mathcal{T}$

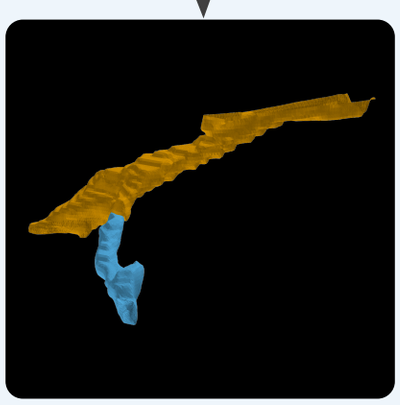
Learned ranking within $\mathcal{T}$

$$\phi(P_e) = \begin{bmatrix} z(f_s(P_e)) \\ -z(\log(1 + f_c(P_e))) \end{bmatrix},$$

$$e^\star = \arg\max_{e \in \mathcal{T}} \theta^\top \phi(P_e)$$

with θ learned on TRAIN

Select intervention $e^\star$



Automatic correction

CC-Search

Paired-contact search and ranked alternatives

H_FIXED single-contact base

$$s_{\mathrm{H}}(P_e) = 10^9 f_{\mathrm{bc}}(P_e) - 10^5 f_{\mathrm{nextra}}(P_e) + 10^2 f_s(P_e) - 10^{-3} f_c(P_e)$$

Bounded contact pairs
Unordered pairs from $\mathcal{C}$; up to B = 45, sampled if more

Generate paired-contact partitions
Both orientations; seeded s-t min-cut; skip conflicting source-sink assignments

Deduplicate/canonicalize
Treat side-swapped partitions as equivalent

Facility location
Select up to K = 5 diverse alternatives; greedy coverage with fixed base

PairLogit
$r(P) = w^\top \psi(P)$; with w learned from opposite-label pairs

H_FIXED base + up to K ranked alternatives
Panel for proofreading or downstream selection

H_FIXED base



Up to K = 5 ranked alternatives

